



RFC-ATF-4

Agent Trust Fabric

Proactive Governance Layer

Anticipatory Veto Protocol · Dynamic Semantic Portability · Structural Shift Detection
Dual Formal Verification · 19 Z3 SMT Proofs UNSAT · 108 Conformance Vectors

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RFC-ATF-4: Agent Trust Fabric – Proactive Governance Layer
Anticipatory Veto Protocol, Structural Shift Detection,
and Dynamic Semantic Portability
Extension to RFC-ATF-1, RFC-ATF-2, and RFC-ATF-3
Version 1.0.0 – OMNIX QUANTUM Open Standard

Abstract

This document specifies the Proactive Governance Layer (PGL) of the Agent Trust Fabric protocol – the fourth RFC in the ATF Open Standard series. RFC-ATF-5 (Cognitive Governance Layer) extends this work.

RFC-ATF-1 answered the identity and delegation question: who authorized this agent, with what authority, and can that be proved offline?

RFC-ATF-2 answered the runtime continuity question: did authority remain valid throughout execution, and was every health degradation event cryptographically attested? RFC-ATF-3 answered the evidence question: where does the resulting artifact go, who can interpret it across organizational boundaries, and can a regulator reconstruct the full chain of custody years later without platform access?

RFC-ATF-4 answers a fourth, structurally harder class of questions that no existing governance framework has formally addressed:

- (1) What happens between governance requests? A system that evaluates drift only when asked can issue receipts under conditions it would have rejected, simply because no request arrived during the adverse period. The Anticipatory Governance Veto Protocol (AGVP) closes this gap by emitting cryptographically signed, database-persisted ProactiveVetoReceipts before any request arrives – anchoring the forensic record to the moment of detection rather than the moment of the next request.
- (2) Is it safe to recalibrate? When the Assumption Validity Monitor detects that a domain has drifted beyond its threshold, two fundamentally different conditions are indistinguishable without additional analysis: sustained excursion from a stable signal composition (recalibration safe) versus structural change in which signals drive governance instability (recalibration unsafe). The Structural Shift Detector (SSD) distinguishes them using the Component Rank Stability Index (CRSI) – a continuous, formally bounded metric proved correct across all real-number inputs.
- (3) Does a receipt issued today remain semantically legitimate in a receiving domain six months from now? A receipt preserves cryptographic integrity but not semantic portability: the meaning of its governance fields may have changed as regulatory contexts evolved, sovereign policies were updated, or operational definitions shifted. The Dynamic Semantic Portability Protocol (DSPP) provides a unilateral, offline-computable Retroactive Semantic Assessment (RSA) that any receiving domain can perform without contacting the originating runtime – $O(1)$ per receipt per domain, no bilateral negotiation required.

Together, AGVP, SSD, and DSPP constitute the Proactive Governance Layer – the governance infrastructure answer to three open problems whose solutions converge on a single architectural insight: governance evidence must exist before the question is asked, the assessment must be topologically aware, and the result must travel independently of the platform that produced it.

Nineteen formal properties are proved by the OMNIX Formal Verification Suite (OMNIX-FVS-1.0) using the Z3 SMT solver: the new invariants

introduced in this RFC plus three foundational invariants from RFC-ATF-1 and RFC-ATF-2 (ATF-INV-001, ATF-INV-004, RGC-INV-004), providing end-to-end arithmetic verification of the complete ATF stack. Three of those properties (ATF-INV-001, ATF-INV-004, RGC-INV-004) are additionally verified by TLA+ model checking, making this specification the first AI governance RFC with dual-methodology formal verification: machine-checkable arithmetic proofs across the continuous input domain (Z3) and state-machine safety proofs across all discrete execution traces (TLA+).

An implementation that complies with RFC-ATF-1, RFC-ATF-2, RFC-ATF-3, and RFC-ATF-4 is designated ATF-PGL-Compliant (Proactive Governance Layer Compliant) – the fourth compliance tier in the ATF stack. RFC-ATF-5 defines the fifth tier: ATF-CGL-Compliant.

Status of This Memo

This is an OMNIX QUANTUM Open Standard, published under the OMNIX Open Governance License v1.0. This document extends RFC-ATF-1, RFC-ATF-2, and RFC-ATF-3 and MUST be read in conjunction with all three. Implementers of RFC-ATF-3 who require anticipatory governance detection, recalibration safety verification, or cross-domain semantic portability SHOULD adopt RFC-ATF-4 as specified herein.

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Acknowledgements

The three open problems addressed in this RFC were crystallized through technical engagement with the broader governance and verification research community.

The detection latency problem and the recalibration topology problem (§2.1, §2.3) emerged from technical dialogue with Reza Zarei, creator of the 3S Silent Authority System, whose questions – "How do you detect that a shift is structural versus a sustained drift pattern before review is triggered?" and "How early can you issue the veto before the next request?" – established the precise research questions that AGVP and SSD answer.

The semantic portability problem (§2.2) was formally articulated by Antonio Socorro in a May 2026 cross-system technical review of RFC-ATF-3: "Where my research diverges is around sealing semantic continuity entirely at receipt creation time. While a receipt may preserve cryptographic integrity, authority lineage, and admissibility state, the receiving domain may still reinterpret operational meaning differently as regulatory conditions evolve, sovereign policies change, runtime contexts shift, or governance semantics diverge over time." This observation is the founding motivation for DSPP.

The dual-methodology formal verification approach (§8) was inspired by the publication of VeriSigil's Z3 proof set (Babatunde, 2026), which

demonstrated the value of machine-checkable arithmetic proofs for governance specifications. OMNIX-FVS-1.0 extends that contribution with 19 proofs – more than four times the VeriSigil count – and combines it with TLA+ model checking that no other published governance specification provides.

Table of Contents

1. Introduction	7
2. Problem Statement: The Proactive Governance Gap	9
2.1. The Detection Latency Problem	9
2.2. The Semantic Portability Problem	11
2.3. The Recalibration Topology Problem	12
2.4. The Proactive Governance Gap (Gap_PG)	13
3. Conventions and Terminology	14
4. Architecture: The Proactive Governance Layer	17
4.1. Position in the ATF Stack	17
4.2. Extension Relationships	18
4.3. Non-Destructive Integration	19
5. Anticipatory Governance Veto Protocol (AGVP)	20
5.1. Two-Layer Veto Architecture	20
5.2. ProactiveVetoReceipt (PVR) Structure	21
5.3. AGVPWatchdog Specification	24
5.4. Signal Telemetry Update Protocol	26
5.5. PVR Lifecycle: ACTIVE and REVOKED	27
5.6. AGVP Invariants (AGV-INV-001-006)	28
6. Structural Shift Detector (SSD)	32
6.1. Signal Topology and Recalibration Safety	32
6.2. Component Rank Stability Index (CRSI) Algorithm ...	34
6.3. Position-Weighting Rationale	37
6.4. StructuralShiftReport Format	38
6.5. Integration with AGVP and AVM	39
6.6. SSD Invariants (SSD-INV-001-003)	40
7. Dynamic Semantic Portability Protocol (DSPP)	42
7.1. Semantic Posture Vector (SPV)	43
7.2. Semantic Distance Unit (SDU)	44
7.3. Temporal Semantic Anchor (TSA)	48
7.4. Semantic Drift Record (SDR)	50
7.5. Retroactive Semantic Assessment (RSA)	52
7.6. Portability Verdicts	55
7.7. DSPP Invariants (DSPP-INV-001-007)	56
8. Formal Verification (OMNIX-FVS-1.0)	60
8.1. Dual Methodology	60
8.2. Z3 SMT Proof Inventory	62
8.3. TLA+ Model-Checked Properties	64
8.4. Machine Reproducibility	64
9. Proactive Governance Layer – Composition	65
9.1. Layer Architecture	65
9.2. Cross-Layer Integration Points	66
9.3. Failure Isolation	67
10. Combined Invariant Summary	68
11. Compliance Designation: ATF-PGL-Compliant	71
11.1. Designation Requirements	71
11.2. Compliance Hierarchy	72
12. Implementation Requirements	73
12.1. AGVP Requirements	73
12.2. SSD Requirements	74
12.3. DSPP Requirements	75
12.4. Formal Verification Requirements	76
13. Security Considerations	77
13.1. ProactiveVetoReceipt Spoofing	77
13.2. Semantic Anchor Replay Attack	78
13.3. CRSI History Flooding	79
13.4. Watchdog Starvation	80
13.5. SDR Chain Forgery	80
13.6. Zero-Discount Delegation Cycle	81
14. Novel Contributions	82
14.1. Anticipatory Governance Veto Receipt (PVR)	82

14.2.	Component Rank Stability Index (CRSI)	83
14.3.	DSPP – O(1) Cross-Domain Assessment Without Negotiation	84
14.4.	Dual-Methodology Formal Verification (TLA+ + Z3 SMT)	84
14.5.	Proactive Governance Gap (Gap_PG) Formal Definition	85
14.6.	ATF-PGL-Compliant – Fourth Compliance Tier	85
15.	Distinction from Prior Art	85
16.	Regulatory Alignment	87
17.	References	88
18.	Appendix A – AGVP Wire Format Reference	90
19.	Appendix B – SDU Computation Reference	92
20.	Appendix C – PGL Compliance Checklist	93
21.	Author's Address	96

1. Introduction

The Agent Trust Fabric protocol stack was designed to address the complete lifecycle of a governed autonomous agent action: from the moment authority is delegated (RFC-ATF-1), through the continuous monitoring of that authority at runtime (RFC-ATF-2), to the long-term preservation and independent verification of the resulting evidence (RFC-ATF-3).

RFC-ATF-1 answered: who authorized this agent, with what authority, and can that be proved offline?

RFC-ATF-2 answered: was authority continuously valid during execution, and is every health event timestamped and signed?

RFC-ATF-3 answered: is the resulting evidence classified, lifecycle-managed, forensically packaged, and independently verifiable years later without platform access?

Each RFC answered its question completely. But three structural problems were deliberately left open – not because they were unknown, but because they require a different class of mechanism than the first three RFCs provide. The first three RFCs operate at request time. RFC-ATF-4 operates in the intervals between requests, across domain boundaries over time, and underneath the metric surfaces that the first three RFCs expose.

This document addresses all three simultaneously.

The Detection Latency Problem: a request-time governance system has no forensic record of adverse conditions that existed before the next request arrived. If a regulatory shock occurs at $T=0$ and no governance request arrives until $T=4h$, a compliant RFC-ATF-3 system would issue receipts at $T=4h$ with no evidence that conditions were already adverse at $T=0$. This is not a bug – it is the structural consequence of reactive governance design. AGVP closes it.

The Semantic Portability Problem: a receipt that is cryptographically valid in the issuing domain may carry a different operational meaning in a receiving domain that has evolved its regulatory interpretation since the receipt was issued. No protocol exists to assess, without bilateral re-negotiation, whether a specific receipt from a specific moment in time is semantically portable to a specific receiving domain in its current state. DSPP closes it.

The Recalibration Topology Problem: when the Assumption Validity Monitor detects that a governance domain has drifted beyond its threshold, the operator faces a binary decision – recalibrate or not – without knowing whether the drift is a sustained excursion from a stable signal composition (recalibration safe) or a fundamental change in which signals drive governance instability (recalibration would embed an obsolete topology as the new baseline). SSD closes it.

These three problems are structurally independent in their motivations. They converge on the same architectural conclusion: governance evidence must precede requests, semantic legitimacy must be computable offline at any future time, and topology changes must be distinguished from excursions before any recalibration action is taken. The Proactive Governance Layer integrates all three.

2. Problem Statement: The Proactive Governance Gap

This section formally states the three open problems and defines the Proactive Governance Gap (Gap_PG) that RFC-ATF-4 closes. The formal structure parallels the Governance Evidence Gap (Gap_GE) defined in RFC-ATF-3 §2.

2.1. The Detection Latency Problem

Define: for a governance domain D with a calibrated AVM baseline, Reactive Veto Coverage (RVC) is the property that every governance request processed during an adverse period carries a `block_reason` in its receipt.

RFC-ATF-1 through RFC-ATF-3 establish RVC at request boundaries. A compliant deployment satisfies: for any request R processed when domain D is in adverse state, $R.\text{receipt.blocked} = \text{True}$.

Define: Proactive Veto Coverage (PVC) is the stronger property that for every interval $[T_{\text{adverse}}, T_{\text{request}})$, where T_{adverse} is the moment adverse conditions were first measurable and T_{request} is the moment of the next governance request, there exists at least one signed, database-persisted governance artifact timestamped within $[T_{\text{adverse}}, T_{\text{adverse}} + \text{AGVP_WATCHDOG_INTERVAL}]$ attesting to the adverse conditions.

Formally:

RVC: $\forall \text{ request } R: \text{domain_is_adverse}(D, R.\text{timestamp}) \rightarrow R.\text{blocked}$
 PVC: $\forall \text{ adverse interval } [t_{\text{start}}, t_{\text{end}}]: \exists \text{ PVR } p: t_{\text{start}} \leq p.\text{timestamp} \leq t_{\text{end}} + \text{AGVP_MIN_INTERVAL} \wedge p.\text{domain} = D \wedge p.\text{status} = \text{ACTIVE}$

RVC does not imply PVC. A deployment can satisfy RVC (blocking every request during the adverse period) while having $\text{Gap_PG} > 0$ (no record of the adverse period between requests).

Example: the Terra/LUNA governance scenario (May 2022). A system satisfying only RVC would have no governance artifacts for the four hours between market open and the first governance-triggered request. A system satisfying PVC would have PVRs timestamped within 60 seconds of the adverse condition onset. For regulated financial systems, the difference is a four-hour forensic gap versus a 60-second forensic gap.

2.2. The Semantic Portability Problem

Define: Semantic Portability (SP) for a receipt R from issuing domain D_{issuer} to receiving domain D_{receiver} is the property that:

$\text{SP}(R, D_{\text{receiver}}) \equiv$ the governance interpretation of $R.\text{fields}$ by D_{receiver} is compatible with their intended meaning at issuance time in D_{issuer}

ATF-INV-006 (RFC-ATF-1) establishes Independent Cryptographic Verifiability (ICV) – any party holding the issuer's public key can verify R 's signature. ICV does not imply SP.

Define: Temporal Semantic Drift (TSD) is the change in D_{receiver} 's interpretation of ATF Core Terms between $R.\text{issued_at}$ and the current

time – resulting from regulatory evolution, sovereign policy updates, or operational definition changes.

Without RFC-ATF-4:

SP is undefined: no protocol exists to assess, without contacting D_issuer, whether R remains semantically portable after TSD.

The only mechanism available is bilateral Semantic Alignment Certificate (SAC) renegotiation (RFC-ATF-3 §5.3), which requires live coordination with the originating runtime and produces a new bilateral artifact – it does not retroactively assess existing receipts.

RFC-ATF-4 §7 (DSPP) closes this gap by enabling any D_receiver to compute $SP(R, D_receiver)$ locally, at any future time, using only public artifacts: the Temporal Semantic Anchor (TSA) sealed at R.issued_at and the public Semantic Drift Record (SDR) chain for D_issuer. No contact with D_issuer required.

2.3. The Recalibration Topology Problem

Define: for a domain D, a Drift Excursion (DE) is the condition where $D.drift_score > D.threshold$ for some continuous interval $[t_{\square}, t_{\square}]$.

Define: a Topological Stability (TS) condition holds for D when the set of signals driving $D.drift_score > D.threshold$ has not materially changed since D's calibration baseline was established.

Two DE conditions require different recalibration policy:

DE $\wedge$ TS: Drift is a sustained excursion from a stable signal composition. Recalibrating to current values is safe: the governance assumption topology is unchanged. The new baseline captures the current operating envelope.

DE $\wedge$ $\neg$ TS: Drift is accompanied by a change in which signals are the primary drivers of governance instability. The calibration frame may be entirely obsolete. Recalibrating to current values embeds an unvalidated topology as the governance baseline. This is semantically equivalent to declaring "current conditions are normal" when the definition of "normal" has changed.

Without RFC-ATF-4, TS is undefined. The AVM detects DE but cannot assess TS. The recalibration decision is therefore unguided by topological evidence.

RFC-ATF-4 §6 (SSD) closes this gap with the Component Rank Stability Index (CRSI) – a formally bounded metric that classifies whether the drift-signal composition of a domain is topologically stable, and therefore whether auto-recalibration is safe.

2.4. The Proactive Governance Gap (Gap_PG)

The Proactive Governance Gap is defined as:

$$Gap_PG = (PVC \cup SP \cup TS) - (ICV \cup RVC)$$

An ATF-FEI-Compliant deployment without RFC-ATF-4 has $Gap_PG > 0$. Specifically:

PVC is undefined: no mechanism exists to issue signed veto evidence between governance requests.

SP is undefined: no mechanism exists to assess semantic portability of existing receipts in evolving receiving domains.

TS is undefined: no mechanism exists to distinguish topological

stability from excursion stability in drift analysis.

RFC-ATF-4 closes Gap_PG by specifying AGVP (§5) for PVC, SSD (§6) for TS, and DSPP (§7) for SP.

3. Conventions and Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174].

This document uses all terms defined in RFC-ATF-1 §2, RFC-ATF-2 §3, and RFC-ATF-3 §3 without redefinition. Additional terms:

Proactive Governance Layer (PGL):

The sixth operational layer of the ATF stack (L6) in the six-plane model used throughout this document (L1-L6), defined by this RFC. Comprises AGVP, SSD, and DSPP operating in concert. RFC-ATF-5 presents a consolidated five-layer model in which RFC-ATF-1's three sub-planes (L1/L2/L3) are treated as a single Layer 1; the PGL maps to Layer 4 in that consolidated view.

ProactiveVetoReceipt (PVR):

An ML-DSA-65 signed, database-persisted governance artifact emitted by the AGVPWatchdog when drift conditions exceed threshold between governance requests. A PVR constitutes a proactive governance block with authority equivalent to a reactive AVM drift block.

AGVPWatchdog:

The background daemon process that polls cached governance signal telemetry at a configurable interval (minimum 30 seconds per AGV-INV-003) and emits PVRs when adverse conditions are detected.

Semantic Posture Vector (SPV):

A hash-identified snapshot of a runtime's current interpretation of the eight ATF Core Terms: `authority_scope`, `execution_boundary`, `delegation_authority`, `context_boundary`, `temporal_authority`, `cross_domain_authority`, `continuity_obligation`, and `forensic_preservation_scope`.

Temporal Semantic Anchor (TSA):

An ML-DSA-65 signed record created at receipt issuance time, capturing the issuing runtime's Semantic Posture Vector at that exact moment. TSAs are public artifacts, independently publishable, and require no bilateral agreement.

Semantic Drift Record (SDR):

An append-only, ML-DSA-65 signed record published by a runtime each time its Semantic Posture Vector changes. SDRs constitute a public, independently verifiable semantic changelog for any ATF-compliant runtime. SDR chains MUST be append-only (DSPP-INV-002).

Semantic Distance Unit (SDU):

A normalized value in $[0.0, 1.0]$ measuring the semantic divergence between two ATF runtime postures for a given governance term. Formally defined in §7.2. SDU = 0.0 indicates identical postures; SDU = 1.0 indicates maximum divergence.

Retroactive Semantic Assessment (RSA):

A pure function over (TSA, receiving_SPV, SDR_chain) that produces an aggregate SDU and a portability verdict for a specific receipt in a specific receiving domain. The RSA is computable locally, requires no communication with the originating runtime, and produces deterministic output for identical inputs (DSPP-INV-005).

Component Rank Stability Index (CRSI):

A continuous $[0.0, 1.0]$ metric measuring whether the dominant drift-signal components for a governance domain have remained consistent across evaluation cycles. $CRSI = 1.0$ indicates perfect topological stability; $CRSI < 0.50$ with sufficient history (≥ 5 cycles per SSD-INV-003) indicates STRUCTURAL_SHIFT.

Structural Shift:

The condition where the composition of signals driving governance drift for a domain has changed, independent of drift magnitude. A STRUCTURAL_SHIFT classification blocks auto-recalibration (SSD-INV-001) even when no active PVR exists (AGV-INV-006).

StructuralShiftReport:

A structured record embedded in every AVMMResult and every PVR (when issued), classifying the domain's current shift state and providing the CRSI value, dominant components, and emerged/receded signals.

ATF-PGL-Compliant:

The compliance designation for implementations that satisfy all invariants of RFC-ATF-1, RFC-ATF-2, RFC-ATF-3, and RFC-ATF-4. ATF-PGL-Compliant implementations provide the complete six-layer ATF governance infrastructure (L1-L6) including proactive evidence, recalibration safety, and cross-domain semantic portability.

Detection Latency:

The maximum elapsed time between the onset of adverse drift conditions and the creation of a signed governance artifact attesting to those conditions. For ATF-PGL-Compliant deployments, detection latency MUST NOT exceed AGVP_WATCHDOG_INTERVAL_SECONDS (default: 60, minimum: 30).

Topological Recalibration Safety:

The property that a domain's auto-recalibration cycle operates only when the signal composition driving drift has remained stable ($CRSI \geq 0.50$ or INSUFFICIENT_DATA). Guaranteed by SSD-INV-001 and AGV-INV-006 acting independently.

4. Architecture: The Proactive Governance Layer

4.1. Position in the ATF Stack

RFC-ATF-4 adds the sixth operational layer to the ATF architecture. The complete stack, with all six layers, is:

L1 – Identity Plane (RFC-ATF-1 §4):

Agent Identity Records (AIR). ML-DSA-65 key registration. Immutable human-principal binding.

L2 – Delegation Plane (RFC-ATF-1 §5):

Delegation Receipts (DR). Trust Lattice. Monotonic Authority Reduction (ATF-INV-001) enforced at every delegation event. Budget ceiling preservation (ATF-INV-004).

L3 – Verification Plane (RFC-ATF-1 §8):

Independent, offline receipt verification. Any party holding the issuer's public key can verify any artifact. ATF-INV-006 (ICV).

L4 – Runtime Continuity Plane (RFC-ATF-2):

Continuity Eligibility Score (CES). Authority Fragmentation Guard (AFG). Runtime Continuity Records (RCR). Reauthorization Challenge (RC). All eight RGC-INV invariants.

L5 – Forensic Evidence Infrastructure (RFC-ATF-3):

Governance Policy Interoperability (GPIL). Evidence lifecycle

(ELR, EAP). Forensic package format (OEP). Authenticated export (FEA). Forensic verification protocol (FVP).

L6 – Proactive Governance Layer (RFC-ATF-4):

Anticipatory Governance Veto Protocol (AGVP): proactive veto before requests arrive. Structural Shift Detector (SSD): recalibration safety. Dynamic Semantic Portability Protocol (DSPP): cross-domain semantic portability without negotiation.

L6 operates orthogonally to L1-L5. It reads from the Assumption Validity Monitor (L4) and the receipt store (L2). It does not modify any artifacts in L1-L5. A Layer 6 failure MUST NOT affect the ability to issue, verify, or archive governance artifacts.

4.2. Extension Relationships

RFC-ATF-4 extends the following prior specifications:

RFC-ATF-1: All six invariants (ATF-INV-001-006) remain unchanged. The Trust Lattice Acyclicity property (RGC-INV-004, jointly owned with RFC-ATF-2) is Z3-proved in OMNIX-FVS-1.0 (§8.2) for all strict-discount three-node delegation cycles.

RFC-ATF-2: All eight invariants (RGC-INV-001-008) remain unchanged. AGV-INV-006 (§5.6) adds a governance-level block to the AVM auto-recalibration cycle (ADR-120), operating additively alongside the existing recalibration safety checks.

RFC-ATF-3: All twenty-six invariants (GPIL, ELR, EAP, OEP, FEA, FVP families) remain unchanged. DSPP TSAs are classified as FORENSIC evidence (ELR-INV-001) and are subject to the HOT retention guarantee (ELR-INV-004). AGVP PVRs are classified as LEGAL evidence.

4.3. Non-Destructive Integration

The PGL is non-destructive with respect to L1-L5. In particular:

- (a) The AGVPWatchdog operates as a daemon thread. Its failure MUST NOT affect request processing. Watchdog errors are logged at WARNING level and the watchdog re-enters its polling loop.
- (b) AGVP signal telemetry updates are decoupled from the veto path. A governance request updates cached signals even when the domain is blocked. This preserves watchdog observability during a block.
- (c) The SSD history ring buffer is an in-process data structure. Its eviction policy (FIFO, max 20 entries) is deterministic and does not affect any persisted artifact.
- (d) DSPP TSA creation adds one ML-DSA-65 signing operation to the receipt issuance path. This operation is non-blocking and MUST fail gracefully: if TSA signing fails, the receipt MUST still be issued with a log entry at ERROR level noting the TSA failure.

5. Anticipatory Governance Veto Protocol (AGVP)

5.1. Two-Layer Veto Architecture

ATF-PGL-Compliant implementations operate a two-layer veto architecture:

Layer 1 – Reactive Veto (ADR-076, existing):

Trigger: A governance request R arrives.

```

    AVM.evaluate(D, R.signals) returns is_valid = False
    and pass_through = False.
  Authority: DRIFT_BLOCK – the request is rejected.
    R.receipt.blocked = True.
    R.receipt.block_reason = "drift_score={x} > threshold={t}"
  Latency: Detection occurs at request time only. No evidence
    exists between requests.

Layer 2 – Anticipatory Veto (ADR-174, RFC-ATF-4):
  Trigger: AGVWatchdog polls cached signals.
    AVM.evaluate(D, cached_signals[D]) returns is_valid =
    False and pass_through = False.
  Authority: ACTIVE PVR – all subsequent requests for D are blocked
    (AGV-INV-001). The PVR exists in the ledger before
    any request arrives.
  Latency: Detection within one watchdog interval (default 60s,
    minimum 30s per AGV-INV-003).

```

Both layers carry equivalent governance authority (AGV-INV-001). The anticipatory layer does not replace the reactive layer – it precedes it. A request arriving at a domain with an ACTIVE PVR finds the block already established, with a timestamp proving the governance system detected the problem before it was asked.

The two-layer architecture preserves forensic completeness:

```

Without Layer 2: record shows [T=4h: REQUEST_BLOCKED]
With Layer 2:   record shows [T=0:00: PVR_ISSUED, T=4h: REQUEST_BLOCKED,
    block_reason references PVR issued at T=0:00]

```

5.2. ProactiveVetoReceipt (PVR) Structure

A ProactiveVetoReceipt is the primary artifact of the AGVP. Its structure is normative; implementations MUST produce PVRs conforming to this schema:

```

pvr_id (string, REQUIRED):
  Canonical identifier. Format: "PVR-{16 uppercase hex characters}".
  Generated using cryptographically random bytes. MUST be globally
  unique within the deployment.

tenant_id (string, REQUIRED):
  The tenant identifier for the affected domain.

domain (string, REQUIRED):
  The governance domain identifier for which the veto is issued.

drift_score (number, REQUIRED):
  The drift score that triggered the anticipatory veto. Value is
  in (threshold, 1.0]. MUST be > drift_threshold at time of
  issuance (AGV-INV-005 precondition).

drift_threshold (number, REQUIRED):
  The threshold value configured for this domain at issuance time.
  Value in (0.0, 1.0).

drift_components (object, REQUIRED):
  A map of signal_name → drift_value for all signals contributing
  to drift_score at assessment time.

structural_shift_class (string, REQUIRED):
  One of: INSUFFICIENT_DATA | STABLE | DRIFT_WITH_INSTABILITY |
  STRUCTURAL_SHIFT. Populated from the SSD StructuralShiftReport
  computed at issuance time. Embedded in the PQC-signed artifact.

signals_at_assessment (object, REQUIRED):
  A snapshot of all monitored signal values at the moment of
  assessment. Provides independent forensic evidence of conditions.

```

`assessment_timestamp` (string, REQUIRED):
 ISO-8601-UTC timestamp with nanosecond resolution. Records the exact moment conditions were assessed by the AGVPWatchdog.

`veto_effective_from` (string, REQUIRED):
 ISO-8601-UTC timestamp. MUST equal `assessment_timestamp`. Represents the start of the governed block window.

`status` (string, REQUIRED):
 One of: ACTIVE | REVOKED. MUST be ACTIVE at issuance. Transitions to REVOKED only via authorized admin action (AGV-INV-002).

`content_hash` (string, REQUIRED):
 Format: "sha256:{64 lowercase hex characters}".
 SHA-256 of the canonical JSON encoding of committed fields (`pvr_id`, `tenant_id`, `domain`, `drift_score`, `drift_threshold`, `signals_at_assessment`, `assessment_timestamp`, `veto_effective_from`, `snapshot_id`). Committed fields constitute the forensic commitment (AGV-INV-004).

`pgc_signature` (string, REQUIRED):
 ML-DSA-65 (FIPS 204) signature over the `content_hash` bytes. Algorithm identifier MUST be "ML-DSA-65".

`pgc_algorithm` (string, REQUIRED):
 MUST be "ML-DSA-65". Any other value MUST be rejected.

`snapshot_id` (string, OPTIONAL):
 Reference to the AVM calibration snapshot active at issuance time.

The `content_hash` computation is deterministic (AGV-INV-004): identical committed field values ALWAYS produce identical `content_hash` values. Post-issuance field modification invalidates the ML-DSA-65 signature.

A compliant PVR is persisted to the `avm_anticipatory_veto_receipts` table (ADR-174) before the watchdog loop returns. Persistence MUST be atomic: a PVR that fails to persist MUST NOT be logged as ISSUED. The database uniqueness constraint on `pvr_id` prevents duplicate emission under concurrent watchdog execution.

5.3. AGVPWatchdog Specification

The AGVPWatchdog is a daemon thread that provides the detection mechanism for Layer 2 of the two-layer veto architecture.

Algorithm (normative):

```

Initialize:
  interval ← AGVP_WATCHDOG_INTERVAL_SECONDS (default: 60)
  assert interval ≥ AGVP_MIN_INTERVAL_SECONDS (30) [AGV-INV-003]
  if interval < 30: raise ValueError

Loop (executes continuously):
  sleep(interval)
  for each domain D with cached signal telemetry:
    result ← AVM.evaluate(D, cached_signals[D])
    if result.is_valid = False AND result.pass_through = False:
      if NOT has_active_pvr(D):
        ssd_report ← SSD.classify(D)
        pvr ← build_pvr(D, result, ssd_report)
        sign(pvr, ML-DSA-65)
        persist(pvr) [uniqueness constraint on pvr_id]
        log(ANTICIPATORY_VETO_ISSUED, pvr_id=pvr.pvr_id, domain=D,
           drift_score=result.drift_score)
      else:
        log(PVR_ALREADY_ACTIVE, domain=D, reason="no duplicate issuance")

```

```

elif drift_subsiding(D) AND has_active_pvr(D):
    log(RECOVERY_CANDIDATE, domain=D, drift=result.drift_score)
    # AGV-INV-002: watchdog MUST NOT revoke – RECOVERY_CANDIDATE is informational only

```

The watchdog MUST NOT issue PVRs for domains with `pass_through = True` (AGV-INV-005). A `pass_through` domain has no calibration baseline; without a baseline, no meaningful adverse condition can be declared.

The watchdog MUST NOT revoke PVRs under any condition (AGV-INV-002). The `RECOVERY_CANDIDATE` log event is informational only. Revocation is exclusively the domain of authorized admin action.

Thread safety: the watchdog loop MUST handle exceptions from `AVM.evaluate()` and `SSD.classify()` without terminating. All exceptions MUST be caught, logged at `ERROR` level, and the loop MUST continue to the next domain.

5.4. Signal Telemetry Update Protocol

The AGVP introduces a critical architectural requirement: signal telemetry MUST be updated by the governance request path even when the domain is blocked.

Rationale: if the watchdog reads only stale telemetry during a block period, it cannot detect drift recovery and cannot produce accurate `RECOVERY_CANDIDATE` events. More importantly, if a domain under a PVR block receives additional requests that update signals toward more adverse conditions, the watchdog needs visibility.

Update rule (normative):

```

On any governance request R for domain D, regardless of whether
D has an ACTIVE PVR:
    cached_signals[D] ← R.signals
    cached_timestamp[D] ← R.arrival_timestamp

```

This update MUST occur before any veto check. The telemetry update is a read operation from the governance perspective – it does not modify the PVR status or the receipt disposition.

A domain's cached signals are the single source of truth for the watchdog. Implementations MUST NOT use stale signals older than $\max(3 \times \text{AGVP_WATCHDOG_INTERVAL_SECONDS}, 300 \text{ seconds})$ for PVR issuance.

5.5. PVR Lifecycle: ACTIVE and REVOKED

A PVR transitions through the following states:

```

PENDING (transient, in-memory only):
    Build and sign phase. Not yet persisted.
    Transition to ACTIVE: on successful database persistence.
    Transition to FAILED: on persistence error.

```

```

ACTIVE (persisted):
    PVR is blocking all governance requests for the affected domain
    (AGV-INV-001). Auto-recalibration is frozen (AGV-INV-006).
    This is the only persistent active state.

```

```

REVOKED (persisted):
    PVR has been revoked by authorized admin action. Block is
    lifted for the affected domain. The PVR record remains in
    the database as a permanent audit trail entry. PVR records
    MUST NOT be deleted after revocation.

```

Revocation requirements:

- (a) Only admin actions authenticated by a valid operator token can revoke a PVR (AGV-INV-002). The AGVPWatchdog MUST NOT revoke.
- (b) A revocation event MUST produce an audit log entry with the revoking admin's identity, the revocation timestamp, and the pvr_id.
- (c) Revocation of a PVR does not create a new PVR. If adverse conditions persist after revocation and the watchdog runs again, a new PVR will be issued with a new pvr_id.

5.6. AGVP Invariants (AGV-INV-001 through AGV-INV-006)

AGV-INV-001 — Anticipatory Authority Equivalence

$\forall$ domain D , $\forall$ governance request R targeting D :
 $\text{has_active_pvr}(D) \rightarrow R.\text{blocked} = \text{True}$

An ACTIVE PVR blocks all governance requests for the affected domain with the same authority as a reactive drift block. The block is not semantically weaker because it was issued proactively. A request blocked by AGV-INV-001 MUST carry a block_reason that references the PVR ID and the PVR's assessment_timestamp.

Z3 proof: prove_agv_inv001_authority_equivalence() in
 omnix_core/formal_verification/agv_invariants_z3.py.
 Negation asserted: has_pvr=True $\wedge$ drift>threshold $\wedge$ blocked=False.
 Result: UNSAT.

AGV-INV-002 — Watchdog Cannot Self-Revoke

The AGVPWatchdog process has no code path that sets a PVR's status to REVOKED. The state transition graph for the watchdog contains no edge from any state to REVOKED.

This invariant is verified by static analysis of the watchdog state machine. The watchdog MAY log RECOVERY_CANDIDATE but MUST NOT modify PVR status.

AGV-INV-003 — Minimum Watchdog Interval

$\forall$ AGVPWatchdog instance W :
 $W.\text{watchdog_interval_seconds} \geq \text{AGVP_MIN_INTERVAL_SECONDS} (30)$

An interval ≤ 0 or < 30 MUST be rejected at initialization with a ValueError. The minimum of 30 seconds prevents excessive AVM load while guaranteeing that adverse conditions are detected within twice the minimum interval in the worst case.

Z3 proof: prove_agv_inv003_minimum_interval() – Negation: system_valid = True $\wedge$ interval < 30. Result: UNSAT.

AGV-INV-004 — PVR Content Hash Commitment

$\forall$ PVR p_1, p_2 :
 $\text{committed_fields}(p_1) = \text{committed_fields}(p_2) \rightarrow$
 $p_1.\text{content_hash} = p_2.\text{content_hash}$

The PVR content_hash is a deterministic function of its committed fields. Post-issuance modification of any committed field invalidates the ML-DSA-65 signature. The content_hash is computed using canonical JSON (keys sorted lexicographically, no whitespace) to ensure determinism across implementations.

Z3 proof: prove_agv_inv004_hash_commitment() – Negation: fields_equal = True $\wedge$ hash_equal = False. Result: UNSAT.

AGV-INV-005 — No Veto Without Baseline

```
emit_pvr(D) → AVM.evaluate(D).is_valid = False
             ^ AVM.evaluate(D).pass_through = False
```

PVRs MUST NOT be emitted for domains without a calibration snapshot. A pass_through domain has no established baseline; a veto without a baseline has no forensic reference point. The watchdog MUST check pass_through = False before emitting any PVR.

Z3 proof: prove_agv_inv005_no_veto_without_baseline() – Negation:
pvr_emitted = True ^ pass_through = True. Result: UNSAT.

AGV-INV-006 — Auto-Recalibration Freeze During Active PVR

```
auto_recalibrate(D) → ¬ has_active_pvr(D)
```

The ADR-120 auto-recalibration cycle MUST NOT recalibrate a domain with an ACTIVE PVR. Recalibrating to drifted conditions destroys the forensic baseline and would allow subsequent requests to proceed under conditions that were classified as adverse. This invariant acts additively with SSD-INV-001: either an active PVR or a STRUCTURAL_SHIFT classification is sufficient to block recalibration.

Z3 proof: prove_agv_inv006_rec calibration_freeze() – Negation:
recalibrate = True ^ has_active_pvr = True. Result: UNSAT.

6. Structural Shift Detector (SSD)

6.1. Signal Topology and Recalibration Safety

The Assumption Validity Monitor (ADR-076) evaluates whether a governance domain's current signal state is within its calibrated envelope. When it is not ($\text{drift_score} > \text{threshold}$), the AVM reports $\text{is_valid} = \text{False}$. The recalibration cycle (ADR-120) periodically updates the calibration baseline to reflect current conditions.

The recalibration decision is safe only when the signal composition that drives drift has remained stable since the baseline was established. Two drift conditions require categorically different treatment:

Condition A – Sustained Drift ($\text{DE} \wedge \text{TS}$):

The drift score exceeds the threshold, but the signals ranked as the primary drivers of drift are the same signals that were dominant in previous evaluation cycles. The governance assumption topology is unchanged. The domain has moved outside its calibrated range, but along a familiar axis.

Safe action: recalibrate. The new baseline captures the current operating envelope with an unchanged signal topology.

Condition B – Structural Shift ($\text{DE} \wedge \neg \text{TS}$):

The drift score exceeds the threshold, AND the signals now ranked as the primary drift drivers have changed. Signals that were dominant are no longer driving drift; signals that were not dominant are now the primary contributors. The calibration frame may be entirely obsolete.

Unsafe action: recalibration would embed the new, unvalidated signal topology as the governance baseline, declaring "this structural change is now normal" without any supervised validation.

Without the SSD, these conditions are indistinguishable from AVM output alone. AVM reports drift_score for Condition A and Condition B identically.

6.2. Component Rank Stability Index (CRSI) Algorithm

The CRSI measures the stability of the dominant drift-signal composition across evaluation cycles for a governance domain.

Parameters (normative):

SSD_TOP_K_COMPONENTS = 3 (the top-K drift drivers considered)
 SSD_MIN_CYCLES = 5 (minimum history cycles for STRUCTURAL_SHIFT)
 SSD_MAX_HISTORY = 20 (ring buffer size)

Position weights by rank:

rank-1 (dominant driver): weight = $K = 3$
 rank-2 (second driver): weight = $K-1 = 2$
 rank-3 (third driver): weight = $K-2 = 1$
 $\text{max_weight} = K \cdot (K+1) / 2 = 6$

Algorithm (normative):

On each call to `AVM.evaluate(D, signals)`:

Step 1: Rank `current_components` descending by drift value.
`current_top_K` $\leftarrow$ top K component names by drift value.

Step 2: Append {components: `current_components`,
 timestamp: `now()`} to `_component_history[D]`.

If `len(_component_history[D]) > SSD_MAX_HISTORY`:
 evict oldest entry (FIFO). [SSD-INV-002]

Step 3: For each history entry `h` in `_component_history[D]`:
`hist_top_K` $\leftarrow$ top K components by drift value in `h`
`weighted_overlap` $\leftarrow$ 0
 for `rank_i` in `range(K)`:
 if `current_top_K[rank_i]` in `hist_top_K`:
 `hist_rank_j` $\leftarrow$ index of `current_top_K[rank_i]` in `hist_top_K`
 `weight` $\leftarrow K - \min(\text{rank_i}, \text{hist_rank_j})$
 `weighted_overlap` $\mathrel{+}= \text{weight}$
`CRSI_h` $\leftarrow \text{weighted_overlap} / \text{max_weight}$

Step 4: `CRSI` $\leftarrow \text{mean}(\text{CRSI_h for all h in history})$.
`cycles_analyzed` $\leftarrow \text{len}(\text{_component_history[D]})$

Step 5: Classification:
 if `cycles_analyzed < SSD_MIN_CYCLES`:
 `shift_class` $\leftarrow$ INSUFFICIENT_DATA
 elif `CRSI >= 0.70`:
 `shift_class` $\leftarrow$ STABLE
 elif `CRSI >= 0.50`:
 `shift_class` $\leftarrow$ DRIFT_WITH_INSTABILITY
 else:
 `shift_class` $\leftarrow$ STRUCTURAL_SHIFT [SSD-INV-003 guarantees
 cycles ≥ 5 here]

CRSI is in $[0.0, 1.0]$ for all valid inputs. Proved by:

CRSI-BOUND-LO (UNSAT): $\text{weighted_overlap} \geq 0 \wedge \text{max_weight} > 0 \rightarrow \text{CRSI} \geq 0$.

CRSI-BOUND-HI (UNSAT): $\text{weighted_overlap} \leq \text{max_weight} \rightarrow \text{CRSI} \leq 1.0$.

Both proofs are in `omnix_core/formal_verification/ssd_invariants_z3.py`.

6.3. Position-Weighting Rationale

The CRSI uses position-weighted Jaccard overlap rather than pure set Jaccard similarity. The rationale is as follows.

Pure set Jaccard treats all K components equally. For K=3:

$\{A, B, C\}$ vs $\{A, X, Y\}$: Jaccard = $1/5 = 0.20$

$\{A, B, C\}$ vs $\{X, B, Y\}$: Jaccard = $1/5 = 0.20$

These two cases are not equivalent for governance stability analysis. In the first case, the dominant driver (rank-1, A) is unchanged. In the second case, only the rank-2 driver (B) is unchanged.

A change in which signal is the dominant driver of governance instability (rank-1) is categorically more significant than a change in the rank-3 driver. Position weighting captures this:

{A,B,C} vs {A,X,Y}: weighted = $(3+0+0)/6 = 0.50$

{A,B,C} vs {X,B,Y}: weighted = $(0+2+0)/6 = 0.33$

The first case scores higher (more stable) because the dominant driver is preserved. The second case scores lower because the dominant driver has changed.

Position weighting satisfies the governance intuition that stability of the dominant signal (rank-1) is the primary indicator of topological continuity.

6.4. StructuralShiftReport Format

Every AVMMResult carries a StructuralShiftReport. When a PVR is emitted by the AGVPWatchdog, the StructuralShiftReport is embedded in the PVR (as the structural_shift_class field and associated metadata). The StructuralShiftReport is ML-DSA-65 signed as part of the PVR's committed fields.

```

ssd_id (string):
    Format: "SSD-{10 uppercase hex characters}".

domain (string):
    The governance domain.

shift_class (string):
    One of: INSUFFICIENT_DATA | STABLE |
    DRIFT_WITH_INSTABILITY | STRUCTURAL_SHIFT.

crsi (number):
    CRSI value in [0.0, 1.0], rounded to 4 decimal places.

cycles_analyzed (integer):
    Number of history entries used in CRSI computation.

dominant_components_current (array of string):
    Top-K component names by current drift value.

dominant_components_baseline (array of string):
    Components appearing in the top-K in ≥ 50% of history entries.

emerged_components (array of string):
    Components present in current_top_K but absent from
    dominant_components_baseline.

receded_components (array of string):
    Components present in dominant_components_baseline but absent
    from current_top_K.

detected_at (string):
    ISO-8601-UTC timestamp.
```

6.5. Integration with AGVP and AVM

The SSD is called by both the AGVPWatchdog (§5.3) and the reactive AVM path on every evaluate() call. This ensures that every governance artifact – whether proactive (PVR) or reactive (AVMResult embedded in a receipt) – carries a fresh StructuralShiftReport.

AGVP integration: the PVR's structural_shift_class field (§5.2) is populated from the SSD result at PVR issuance time. This makes

the topology classification part of the PQC-signed, persisted artifact.

Auto-recalibration integration: the recalibration cycle (ADR-120) MUST consult SSD before executing. The check is:

```
if has_active_pvr(D):
    skip recalibration [AGV-INV-006]
elif SSD.classify(D).shift_class == STRUCTURAL_SHIFT:
    skip recalibration [SSD-INV-001]
else:
    execute recalibration
```

Both guards are independent. A domain can be STRUCTURAL_SHIFT without an ACTIVE PVR (if drift magnitude has recovered but topology has changed). SSD-INV-001 handles this case; AGV-INV-006 handles the ACTIVE PVR case.

6.6. SSD Invariants (SSD-INV-001 through SSD-INV-003)

SSD-INV-001 — Structural Shift Blocks Auto-Recalibration

$\text{auto_recalibrate}(D) \rightarrow \text{SSD.classify}(D).\text{shift_class} \neq \text{STRUCTURAL_SHIFT}$

Independent of and additive to AGV-INV-006. A STRUCTURAL_SHIFT classification blocks auto-recalibration even when no ACTIVE PVR exists for the domain. This invariant prevents the AVM from adopting an unvalidated signal topology as a new governance baseline under the cover of routine recalibration.

Z3 proof: $\text{prove_ssd_inv001_shift_blocks_recalibration}()$ – Negation: $\text{recalibrate} = \text{True} \wedge \text{structural_shift} = \text{True}$. Result: UNSAT.

SSD-INV-002 — Component History Ring Buffer Is Append-Only

$\forall \text{ history entry } h \in \text{_component_history}[D]: h \text{ is not modified or selectively deleted. Eviction is FIFO only (ring buffer size limit = SSD_MAX_HISTORY = 20).}$

The chronological integrity of the topology record MUST be preserved. Selective deletion – removing specific entries to alter the CRSI result – is forbidden. The eviction policy is strictly first-in, first-out based on insertion order.

SSD-INV-003 — STRUCTURAL_SHIFT Requires Minimum History

$\text{SSD.classify}(D).\text{shift_class} = \text{STRUCTURAL_SHIFT} \rightarrow \text{SSD.classify}(D).\text{cycles_analyzed} \geq \text{SSD_MIN_CYCLES} (5)$

A STRUCTURAL_SHIFT verdict MUST NOT be issued with fewer than 5 evaluation cycles in the history. A single-cycle or low-history verdict is statistically unreliable. Domains with fewer than SSD_MIN_CYCLES evaluation cycles receive INSUFFICIENT_DATA regardless of CRSI value.

Z3 proof: $\text{prove_ssd_inv003_minimum_history}()$ – Negation: $\text{cycles} < 5 \wedge \text{structural_shift} = \text{True}$. Result: UNSAT.

7. Dynamic Semantic Portability Protocol (DSPP)

The DSPP enables any ATF-compliant receiving domain to perform a deterministic, offline assessment of whether a governance receipt issued in a different domain at a different point in time remains semantically portable – meaning that the governance fields of the receipt carry the same operational meaning in the receiving domain's current regulatory context as they carried in the issuing domain's context at issuance time.

The DSPP achieves this without any communication with the originating runtime. The assessment is a pure function of three public artifacts:

1. The Temporal Semantic Anchor (TSA): sealed at issuance time by the originating runtime, capturing its semantic posture.
2. The Semantic Drift Record (SDR) chain: the public, append-only changelog of the originating runtime's posture evolution.
3. The receiving domain's current Semantic Posture Vector (SPV): locally available to the receiving domain.

The DSPP RSA scales to arbitrarily many receiving domains with $O(1)$ computation per receipt per domain.

7.1. Semantic Posture Vector (SPV)

An SPV is a structured snapshot of a runtime's current interpretation of the eight ATF Core Terms. ATF Core Terms are the governance terms whose operational meaning is deliberately left sovereign by the ATF protocol (within GPIL bounds, RFC-ATF-3 §5).

The eight ATF Core Terms are:

authority_scope:	what actions are within a delegated agent's authority
execution_boundary:	the boundaries within which execution is considered governed
delegation_authority:	what authority delegation entails
context_boundary:	the scope of execution context
temporal_authority:	how temporal constraints bind authority
cross_domain_authority:	conditions for cross-domain trust
continuity_obligation:	what constitutes continuity of authority
forensic_preservation_scope:	what evidence must be preserved

For each Core Term T , the SPV carries a `SemanticTermEntry` with:

```
content_hash (string):
  SHA-256 of canonical encoding. If two entries have identical
  content_hash, their SDU is 0.0 (hash-equivalence shortcut).
boundary_conditions (array of string):
  The conditions under which  $T$  applies.
operational_scope (array of string):
  The operational domains within which  $T$ 's definition holds.
regulatory_anchors (array of string):
  The regulatory references that inform  $T$ 's definition.
last_updated_at (string):
  ISO-8601-UTC timestamp of last modification.
```

An SPV carries a `spv_hash` computed as SHA-256 over the canonical JSON of all eight `SemanticTermEntry` records.

7.2. Semantic Distance Unit (SDU)

The SDU is the fundamental metric of DSPP. It measures the normalized semantic divergence between two `SemanticTermEntry` records for the same ATF Core Term.

Definition (normative):

```
SDU(term_a, term_b):
  if term_a.content_hash == term_b.content_hash:
    return 0.0 [hash-equivalence shortcut]

  boundary_distance ← |Δ(boundary_conditions)| /
    |union(boundary_conditions)|
```

where $|\Delta(S, S')| = |S \setminus S'|$ (symmetric difference size)
 and $|\text{union}(S, S')| = |S \cup S'|$

$$\text{scope_distance} \leftarrow 1.0 - \frac{|\text{operational_scope_a} \cap \text{operational_scope_b}|}{|\text{operational_scope_a} \cup \text{operational_scope_b}|}$$

(Jaccard distance)

$$\text{regulatory_distance} \leftarrow \frac{|\Delta(\text{regulatory_anchors})|}{|\text{union}(\text{regulatory_anchors})|}$$

$$\text{SDU} \leftarrow 0.40 \cdot \text{boundary_distance} + 0.35 \cdot \text{scope_distance} + 0.25 \cdot \text{regulatory_distance}$$

Weight rationale:

boundary_conditions (0.40, highest weight):
 Boundary conditions are the operational guardrails of a governance term. A change in when a term applies is the most impactful dimension of semantic divergence.

operational_scope (0.35, middle weight):
 Scope defines which operational contexts are governed by the term. Scope divergence changes what is covered.

regulatory_anchors (0.25, lowest weight):
 Regulatory references provide the normative grounding. They are important but often evolve without changing operational meaning (e.g., renumbered provisions).

The weights form a convex combination: $0.40 + 0.35 + 0.25 = 1.00$.
 Proved by SDU-WSUM (UNSAT).

Formal bounds (both Z3-proved, UNSAT):

$\text{SDU} \in [0.0, 1.0]$ for all valid inputs.

The weight constants are structural constants of the DSPP specification (DSPP-INV-007a and DSPP-INV-007b). Implementations MUST NOT accept runtime configuration that alters these weights. A new RFC is required to change them.

7.3. Temporal Semantic Anchor (TSA)

A TSA is created by the originating runtime at the moment a governance receipt is issued. It seals the issuing runtime's semantic posture at the exact moment of issuance.

Structure (normative):

tsa_id (string, REQUIRED):
 Format: "TSA-{16 uppercase hex characters}".

receipt_id (string, REQUIRED):
 The identifier of the governed receipt that this TSA anchors.

issuing_runtime_id (string, REQUIRED):
 The identifier of the originating runtime.

spv_hash (string, REQUIRED):
 SHA-256 of the canonical JSON encoding of semantic_term_registry.
 This is the cryptographic commitment to the posture at issuance.

semantic_term_registry (object, REQUIRED):
 A map of ATF Core Term name $\rightarrow$ SemanticTermEntry for all eight Core Terms. This is the complete semantic posture at issuance.

issued_at (string, REQUIRED):
 ISO-8601-UTC timestamp with nanosecond resolution. MUST equal

the receipt's issuance timestamp.

content_hash (string, REQUIRED):
SHA-256 of canonical JSON of committed TSA fields.

pqc_signature (string, REQUIRED):
ML-DSA-65 signature over content_hash bytes.

A TSA is a public artifact. It MUST be independently publishable without any access restriction. Any receiving domain that can retrieve a TSA can perform an RSA without contacting the issuing runtime (DSPP-INV-004).

TSA creation MUST be atomic with receipt issuance. If TSA creation fails, the receipt MUST still be issued with an ERROR log entry. A receipt without a TSA has reduced portability assurance.

TSA Immutability (DSPP-INV-001):
A TSA is sealed at issuance and MUST NOT be modified.
Its content_hash commits all fields at issuance time.

7.4. Semantic Drift Record (SDR)

A Semantic Drift Record is published by a runtime each time its Semantic Posture Vector changes. SDRs constitute the public semantic changelog for the issuing runtime.

Structure (normative):

sdr_id (string, REQUIRED):
Format: "SDR-{16 uppercase hex characters}".

runtime_id (string, REQUIRED):
The issuing runtime.

previous_spv_hash (string, REQUIRED):
The spv_hash of the SPV prior to this change.
For the initial SDR (baseline), this is the genesis sentinel:
"00".
(64 hexadecimal zeros – the SHA-256 length. Implementations MUST reject a genesis sentinel of any other length.)

new_spv_hash (string, REQUIRED):
The spv_hash of the updated SPV.

changed_terms (array of string, REQUIRED):
The ATF Core Terms that changed in this update.

drift_direction (string, REQUIRED):
One of: MORE_RESTRICTIVE | LESS_RESTRICTIVE | LATERAL.
MORE_RESTRICTIVE: the new posture is narrower (higher SDU implies receiving domains that were compatible may now require re-assessment).
LESS_RESTRICTIVE: the new posture is broader.
LATERAL: the posture changed in a way not captured by the restrictiveness axis.

change_rationale (string, REQUIRED):
A human-readable description of the reason for the posture change.
MUST be present. An empty string is not a valid rationale.

effective_from (string, REQUIRED):
ISO-8601-UTC timestamp when the new posture takes effect.

pqc_signature (string, REQUIRED):
ML-DSA-65 signature.

SDR Append-Only Invariant (DSPP-INV-002):
An SDR entry MUST NOT be modified or deleted after issuance.
The SDR chain for a runtime is an append-only public log. This

guarantees that any domain computing an RSA over the SDR chain obtains the complete, unmodified history of semantic evolution.

7.5. Retroactive Semantic Assessment (RSA)

The RSA is the core computation of the DSPP. It produces a deterministic portability verdict for a specific receipt in a specific receiving domain.

Inputs:

receipt_id: the receipt to assess
receiving_runtime_id: the receiving domain

Algorithm (normative):

1. $TSA \leftarrow \text{retrieve}(\text{receipt.tsa_id})$
[from public artifact store – no contact with issuing runtime]
2. $\text{sdr_chain} \leftarrow \text{retrieve_all_sdrs}(TSA.\text{issuing_runtime_id})$
[public SDR chain – ordered by effective_from ascending]
3. $\text{receiving_spv} \leftarrow \text{get_current_spv}(\text{receiving_runtime_id})$
[local knowledge – receiving domain's own posture]
4. For each ATF Core Term T in {authority_scope, execution_boundary, delegation_authority, context_boundary, temporal_authority, cross_domain_authority, continuity_obligation, forensic_preservation_scope}:

$\text{originating_entry} \leftarrow TSA.\text{semantic_term_registry}[T]$
 $\text{receiving_entry} \leftarrow \text{receiving_spv}.\text{registry}[T]$
 $\text{term_sdu}[T] \leftarrow \text{SDU}(\text{originating_entry}, \text{receiving_entry})$
5. $\text{aggregate_sdu} \leftarrow \text{mean}(\text{term_sdu.values}())$
 $\text{portability_confidence} \leftarrow 1.0 - \text{aggregate_sdu}$
6. $\text{verdict} \leftarrow \text{classify_verdict}(\text{aggregate_sdu})$
[§7.6 – DSPP-INV-007a and DSPP-INV-007b guarantee totality and exclusivity]
7. Return $\text{RSAResult}(\text{receipt_id}, \text{receiving_runtime_id}, \text{aggregate_sdu}, \text{portability_confidence}, \text{verdict}, \text{term_sdu}, \text{assessed_at}=\text{now}(), \text{tsa_id}=TSA.\text{tsa_id})$

RSA is a pure function (DSPP-INV-005): identical inputs produce identical aggregate_sdu and identical verdict. Two independent receiving domains computing RSA for the same receipt against the same SPV will always produce the same result.

The RSA MUST be computable with no network call to the originating runtime (DSPP-INV-004). An implementation that requires originating runtime availability for RSA computation is non-conformant.

RSA complexity: $O(K)$ where $K = |\text{ATF Core Terms}| = 8$. This is $O(1)$ with respect to the number of receiving domains. Semantic portability assessment scales to arbitrarily many receiving domains without coordination overhead.

7.6. Portability Verdicts

The RSA produces one of four portability verdicts based on aggregate_sdu. The verdict thresholds are structural constants of the DSPP specification (DSPP-INV-007a and DSPP-INV-007b):

SEMANTICALLY_PORTABLE [0.00, 0.10):

The receiving domain's current posture for all assessed terms is sufficiently aligned with the originating posture at issuance.

The receipt may be used without semantic qualification.

DRIFT_ACKNOWLEDGED [0.10, 0.40):
Moderate semantic divergence exists. The receiving domain **SHOULD** apply contextual qualification when using this receipt. The receipt remains admissible but carries semantic drift.

DRIFT_CRITICAL [0.40, 0.70):
Significant semantic divergence. The receiving domain **MUST** document the divergence before relying on this receipt in any regulatory context. Human review is **RECOMMENDED**.

SEMANTICALLY_INCOMPATIBLE [0.70, 1.00]:
The receiving domain's interpretation of the governed terms is sufficiently divergent that the receipt cannot be assumed to carry its original semantic content. The receipt **MAY** still be cryptographically valid (ICV holds), but its operational meaning in the receiving domain is unreliable.

DSPP-INV-006 – Incompatibility Propagation:
A **SEMANTICALLY_INCOMPATIBLE** verdict on a Delegation Receipt propagates to all receipts that descend from that DR in the receiving domain's chain assessment.

7.7. DSPP Invariants (DSPP-INV-001 through DSPP-INV-007)

DSPP-INV-001 — TSA Immutability

A TSA is sealed at receipt issuance and **MUST NOT** be modified. Its content_hash commits all fields at issuance time.

DSPP-INV-002 — SDR Append-Only

SDR entries **MUST NOT** be modified or deleted after publication. SDR chains are append-only public logs.

DSPP-INV-003 — RSA Uses MORE_RESTRICTIVE Governing Posture

When the drift_direction of the most recent SDR is **MORE_RESTRICTIVE**, the RSA **MUST** use the more restrictive boundary interpretation. This is the conservative default: in semantic ambiguity, the receiving domain assumes the stricter interpretation.

DSPP-INV-004 — No Semantic Negotiation

RSA computation requires no communication with the originating runtime. An implementation that requires originating runtime availability for RSA computation is non-conformant.

DSPP-INV-005 — RSA Verdict Determinism

$\forall$ (TSA, receiving_SPV, SDR_chain): RSA is a pure function.
same inputs $\rightarrow$ same aggregate_sdu $\rightarrow$ same verdict

Z3 proof: prove_dspp_inv005_rsa_determinism() – Negation:
same_inputs = True $\wedge$ sdu_A $\neq$ sdu_B. Result: UNSAT.

DSPP-INV-006 — Incompatibility Propagation

A **SEMANTICALLY_INCOMPATIBLE** verdict on a DR in a delegation chain propagates to all receipts descending from that DR in the receiving domain's assessment.

DSPP-INV-007 (a + b) — Structural Threshold Constants

The verdict thresholds (0.10, 0.40, 0.70) and SDU weights (0.40, 0.35, 0.25) are structural constants of the DSPP specification. They **MUST NOT** be overridden by runtime configuration. Changing them requires a new RFC.

Two sub-properties, both Z3-proved:

DSPP-INV-007a – Threshold Partition Totality:

$\forall$ aggregate_sdu $\in [0, 1]$: exactly one verdict class applies.

Z3: Negation: sdu $\in [0, 1] \wedge \neg(\text{PORTABLE} \vee \text{ACKNOWLEDGED} \vee \text{CRITICAL} \vee \text{INCOMPATIBLE})$. Result: UNSAT.

DSPP-INV-007b – Threshold Partition Exclusivity:

$\forall$ aggregate_sdu $\in [0, 1]$: at most one verdict class applies.

Z3: Negation: sdu satisfies two verdict conditions simultaneously.
Result: UNSAT.

8. Formal Verification (OMNIX-FVS-1.0)

8.1. Dual Methodology

OMNIX RFC-ATF-4, combined with the TLA+ properties established in RFC-ATF-1 and RFC-ATF-2, constitutes the first AI governance protocol specification with dual-methodology formal verification.

The two methodologies are strictly complementary:

TLA+ Model Checking:

Scope: safety properties across all discrete execution traces.

Coverage: Monotonic Authority Reduction (MAR), delegation chain acyclicity (structural, non-arithmetic), RC expiry HALT propagation, temporal authority bounds.

Tool: TLA+ with TLC model checker.

Specification: docs/formal/ATF-TLA-SPEC.tla (included in Zenodo).

Properties proved: 5 (MAR, Acyclicity, ChainRoot, Immutability, RCEXpiry).

Z3 SMT Arithmetic:

Scope: correctness of arithmetic-defined invariants across the continuous input domain.

Coverage: all arithmetic invariants in RFC-ATF-1/2/4 – budget bounds, threshold constraints, weight sums, CRSI bounds, classification totality, RSA determinism.

Tool: Z3 version 4.x (de Moura & Bjørner, 2008).

Suite: OMNIX-FVS-1.0 in omnix_core/formal_verification/ (5 modules).

Properties proved: 19 (all UNSAT).

Why both methodologies are necessary:

TLA+ cannot directly prove arithmetic facts about real-valued metrics (CRSI, SDU, drift_score). Z3 cannot directly prove properties over all possible state machine execution traces. A protocol specification that relies on both discrete state-machine safety and continuous arithmetic correctness requires both tools.

No other published AI governance specification applies both.

8.2. Z3 SMT Proof Inventory

The following 19 proofs constitute OMNIX-FVS-1.0. All return UNSAT.

A SAT result or UNKNOWN result means the invariant does not hold as stated; the prover would provide a counterexample.

Invariant ID	Negation Asserted	Result
ATF-INV-001	child_budget > parent_budget	UNSAT
ATF-INV-004	child_budget > 100 (parent ≤ 100, MAR)	UNSAT
RGC-INV-004	bB < bA ∧ bC < bB ∧ bA < bC (3-node cycle)	UNSAT
AGV-INV-001	pvr_active ∧ drift > thr ∧ ¬blocked	UNSAT
AGV-INV-003	system_valid ∧ interval < 30	UNSAT
AGV-INV-004	fields_equal ∧ hash_differs	UNSAT

■ AGV-INV-005	■ $pvr_emitted \wedge pass_through = True$	■ UNSAT
■ AGV-INV-006	■ $recalibrate \wedge has_active_pvr$	■ UNSAT
■ CRSI-BOUND-LO	■ $overlap \geq 0 \wedge max > 0 \wedge crsi < 0$	■ UNSAT
■ CRSI-BOUND-HI	■ $overlap \leq max \wedge max > 0 \wedge crsi > 1$	■ UNSAT
■ CRSI-CLASS-TOT	■ $crsi \in [0,1] \wedge no\ class\ applies$	■ UNSAT
■ SSD-INV-001	■ $recalibrate \wedge structural_shift = True$	■ UNSAT
■ SSD-INV-003	■ $cycles < 5 \wedge structural_shift = True$	■ UNSAT
■ SDU-BOUND-LO	■ $b,s,r \in [0,1] \wedge sdu < 0$	■ UNSAT
■ SDU-BOUND-HI	■ $b,s,r \in [0,1] \wedge sdu > 1$	■ UNSAT
■ SDU-WSUM	■ $0.40 + 0.35 + 0.25 \neq 1.0$	■ UNSAT
■ DSPP-INV-005	■ $same_inputs \wedge sdu_A \neq sdu_B$	■ UNSAT
■ DSPP-INV-007a	■ $sdu \in [0,1] \wedge no\ verdict\ class\ applies$	■ UNSAT
■ DSPP-INV-007b	■ $sdu\ simultaneously\ in\ two\ verdict\ classes$	■ UNSAT

Total: 19 proofs · 19 UNSAT · 0 SAT · 0 UNKNOWN · runtime < 200ms

The table is ordered: RFC-ATF-1/2 invariants first (rows 1-3), AGVP invariants (rows 4-8), SSD/CRSI invariants (rows 9-13), DSPP/SDU invariants (rows 14-19). This ordering reflects the layered architecture of the ATF stack.

8.3. TLA+ Model-Checked Properties

The TLA+ specification (docs/formal/ATF-TLA-SPEC.tla) model-checks the following five properties across all reachable states of the ATF delegation state machine:

MARInvariant:

In every reachable state, every Delegation Receipt has $authority_budget \leq$ its parent's $authority_budget$.

AcyclicityInvariant:

The delegation graph has no directed cycle in any reachable state. (Structural property – complements the Z3 arithmetic proof of RGC-INV-004 which proves the strict-discount case.)

ChainRootInvariant:

Every delegation chain traces to exactly one human principal AIR.

ImmutabilityInvariant:

Once a Delegation Receipt is issued, its $content_hash$ and $pqc_signature$ fields are never modified.

RCExpiryHalt:

If a Reauthorization Challenge expires (RC_TTL exceeded), the runtime transitions to HALT and HALT propagates to all child sessions.

8.4. Machine Reproducibility

The OMNIX-FVS-1.0 proof suite is machine-reproducible:

```
python -m omnix_core.formal_verification.run_all
```

JSON output mode:

```
python -m omnix_core.formal_verification.run_all --json \
--output proof_report.json
```

The resulting proof_report.json contains for each invariant: invariant_id, result (UNSAT/SAT/UNKNOWN), proved (boolean), elapsed_ms, description, negation_asserted, adr_reference, rfc_reference, and model_counterexample (null if UNSAT).

The OMNIX-FVS-1.0 suite itself is governed by three meta-invariants defined in ADR-177 [ADR-177]:

FVS-INV-001 – All proofs return UNSAT. Any SAT or UNKNOWN result is a suite failure, not just a proof failure.

FVS-INV-002 – No UNKNOWN results. A Z3 timeout violates the soundness assumption of the proof methodology.

FVS-INV-003 – Suite completes in under 10 seconds on reference hardware (Python 3.11, z3-solver 4.x, single core).

FVS-INV-001 through FVS-INV-003 are verified by tests/test_formal_verification.py (class TestFullSuite).

pytest integration:

```
pytest tests/test_formal_verification.py [19 assertions]
```

Dependencies: z3-solver (pip install z3-solver). No platform imports. Reproducible in any Python 3.11+ environment with z3-solver installed.

9. Proactive Governance Layer — Composition

9.1. Layer Architecture

The full six-layer ATF stack, with all cross-layer interactions, is:

L6 ■■■ Proactive Governance Layer (RFC-ATF-4)

- AGVP: anticipatory vetoes before requests arrive
- SSD: recalibration safety under topology change
- DSPP: semantic portability across domain boundaries
- ↓ reads: cached signal telemetry (L4), receipt store (L2)
- ↓ writes: PVR table, TSA table, SDR chain

L5 ■■■ Forensic Evidence Infrastructure (RFC-ATF-3)

- GPIL: policy interoperability contracts
- ELR/EAP: evidence lifecycle and archive pipeline
- OEP/FEA/FVP: forensic packages and verification
- ↓ reads: receipt store (L2), RCR state (L4)
- ↓ writes: COLD blocks, OEP packages, evidence custody log

L4 ■■■ Runtime Continuity Plane (RFC-ATF-2)

- CES: Continuity Eligibility Score
- AFG: Authority Fragmentation Guard
- RCR: Runtime Continuity Records
- RC: Reauthorization Challenge

L2/L3 ■ Delegation and Verification Planes (RFC-ATF-1)

- DR: Delegation Receipts
- TL: Trust Lattice
- Offline Verification (ATF-INV-006)

L1 ■■■ Identity Plane (RFC-ATF-1)

- AIR: Agent Identity Records
- ML-DSA-65 key registration
- Immutable human-principal binding (ATF-INV-002)

9.2. Cross-Layer Integration Points

From	To	Mechanism	Invariant(s)
L6	L2	PVR blocks DR issue	AGV-INV-001
L6	L4	AGVP blocks recal.	AGV-INV-006
L6	L4	SSD blocks recal.	SSD-INV-001
L6	L6	SSD class in PVR	structural_shift_class
L6	L2	TSA per DR	DSPP-INV-001
L6	L5	TSA as FORENSIC ev.	ELR-INV-001

L5	L6	HALT triggers PVR	AGV-INV-005 pre-check	
L4	L6	Signal telemetry	§5.4 update protocol	

9.3. Failure Isolation

PGL failure modes MUST be isolated from governance receipt issuance:

AGVPWatchdog failure:

The watchdog MUST catch all exceptions and re-enter the polling loop. A crashed watchdog reduces the PVC guarantee but MUST NOT affect RVC (Layer 1 reactive vetoes remain fully operative).

SSD classify() failure:

If SSD.classify() raises an exception, the AVMMResult MUST be returned with shift_class = INSUFFICIENT_DATA rather than propagating the exception. Receipt issuance is unaffected.

TSA signing failure:

If TSA creation fails during receipt issuance, the receipt MUST still be issued. An ERROR log entry MUST be written noting the failure. The receipt is issued without DSPP portability assurance.

SDR publication failure:

SDR publication is a background operation. Failure MUST be logged at ERROR level and retried. Runtime operations are unaffected by SDR publication failures.

10. Combined Invariant Summary

The following table lists all twenty-three invariant properties defined in RFC-ATF-4 – six canonical named identifiers per the AGV family, six per the SSD/CRSI family, and eleven per the DSPP/SDU family – with their sections, proof method, and the open problem each closes.

AGVP Invariants:

Invariant	Description	§	Proof	Gap
AGV-INV-001	Anticipatory Authority Equivalence	5.6	UNSAT	PVC
AGV-INV-002	Watchdog Cannot Self-Revoke	5.6	State	PVC
AGV-INV-003	Minimum Watchdog Interval ($\geq 30s$)	5.6	UNSAT	PVC
AGV-INV-004	PVR Content Hash Commitment	5.6	UNSAT	PVC
AGV-INV-005	No Veto Without Calibration Baseline	5.6	UNSAT	PVC
AGV-INV-006	Auto-Recalibration Freeze During PVR	5.6	UNSAT	TS

SSD Invariants:

Invariant	Description	§	Proof	Gap
CRSI-BOUND-LO	CRSI ≥ 0.0 for all valid inputs	6.2	UNSAT	TS
CRSI-BOUND-HI	CRSI ≤ 1.0 for all valid inputs	6.2	UNSAT	TS
CRSI-CLASS-TOT	CRSI classification is total (covers)	6.2	UNSAT	TS
SSD-INV-001	STRUCTURAL_SHIFT Blocks Auto-Recal.	6.6	UNSAT	TS
SSD-INV-002	History Ring Buffer Is Append-Only	6.6	State	TS
SSD-INV-003	STRUCTURAL_SHIFT Requires ≥ 5 Cycles	6.6	UNSAT	TS

DSPP Invariants:

Invariant	Description	§	Proof	Gap
SDU-BOUND-LO	SDU ≥ 0.0 for all valid inputs	7.2	UNSAT	SP
SDU-BOUND-HI	SDU ≤ 1.0 for all valid inputs	7.2	UNSAT	SP
SDU-WSUM	Weights sum to 1.0 (convex combination)	7.2	UNSAT	SP
DSPP-INV-001	TSA Immutability	7.7	Struct	SP
DSPP-INV-002	SDR Append-Only	7.7	Struct	SP

DSPP-INV-003	RSA Uses MORE_RESTRICTIVE Posture	7.7	Logic	SP	
DSPP-INV-004	No Semantic Negotiation Required	7.7	Logic	SP	
DSPP-INV-005	RSA Verdict Determinism	7.7	UNSAT	SP	
DSPP-INV-006	Incompatibility Propagation	7.7	Logic	SP	
DSPP-INV-007a	Threshold Partition Totality	7.7	UNSAT	SP	
DSPP-INV-007b	Threshold Partition Exclusivity	7.7	UNSAT	SP	

Proof column key:

- UNSAT – formally proved by Z3 SMT solver (OMNIX-FVS-1.0); all return UNSAT (unsatisfiable negation), confirming the property holds for all valid inputs
- State – verified by state-machine analysis (exhaustive code-path inspection confirming no reachable state violates the invariant; not amenable to Z3 arithmetic encoding)
- Struct – verified by structural analysis (data model enforces the property by construction; no runtime check needed)
- Logic – verified by logical argument (property follows directly from definition; Z3 encoding would be circular)

Gap column key (open problem closed by each invariant):

- PVC – Proactive Veto Coverage: closes the detection-latency gap (Gap_PG §2.4); governance evidence is cryptographically attested before the request it governs arrives
- TS – Topology Safety: closes the recalibration topology problem (§2.3); STRUCTURAL_SHIFT is formally distinguishable from a sustained drift excursion, making recalibration safety a decidable property
- SP – Semantic Portability: closes the semantic portability problem (§2.2); RSA verdict is deterministic and domain-independent for all valid inputs with no pairwise negotiation required

All invariant_id values in this table are identical to the invariant_id fields in omnix_core/formal_verification/proof files and the run_all.py output.

11. Compliance Designation: ATF-PGL-Compliant

11.1. Designation Requirements

An implementation claims the ATF-PGL-Compliant designation by satisfying all requirements of RFC-ATF-1, RFC-ATF-2, RFC-ATF-3, and this RFC.

ATF-PGL-Compliant REQUIRES, in addition to all ATF-FEI-Compliant requirements:

AGVP

- (a) AGVPWatchdog deployed as a persistent daemon with interval ≥ 30 s
- (b) PVR issued and persisted within one watchdog interval of adverse condition onset when cached signals are current
- (c) PVRs carry ML-DSA-65 signatures over committed fields
- (d) PVR status transitions: ACTIVE only at issuance, REVOKED only by authenticated admin action
- (e) Watchdog: no code path sets PVR.status = REVOKED
- (f) Signal telemetry updated by request path even when domain blocked
- (g) PVRs not emitted for pass_through domains
- (h) Auto-recalibration cycle consults has_active_pvr() before execution

SSD

- (i) CRSI computed on every AVMM.evaluate() call
- (j) StructuralShiftReport embedded in every AVMMResult and every PVR
- (k) Auto-recalibration blocked when shift_class = STRUCTURAL_SHIFT
- (l) Component history ring buffer: FIFO eviction only, no selective deletion; max 20 entries per domain

(m) STRUCTURAL_SHIFT verdict issued only with `cycles_analyzed` ≥ 5

DSPP

- (n) TSA created at receipt issuance for every Delegation Receipt
- (o) TSA carries ML-DSA-65 signature; TSAs publicly accessible
- (p) SDRs published on every SPV change with non-empty rationale
- (q) SDR chain is append-only; no SDR is modified or deleted
- (r) RSA computation does not require live contact with originating runtime
- (s) Verdict thresholds (0.10, 0.40, 0.70) not overridable at runtime
- (t) SDU weights (0.40, 0.35, 0.25) not overridable at runtime

Formal Verification

- (u) OMNIX-FVS-1.0 (or equivalent) executes without UNKNOWN or SAT results for all 19 listed invariants (§8.2)

11.2. Compliance Hierarchy

The ATF compliance designations form a strict hierarchy:

```

ATF-Compliant      (RFC-ATF-1 only)
  ↓ extends
ATF-RGC-Compliant  (+ RFC-ATF-2)
  ↓ extends
ATF-FEI-Compliant  (+ RFC-ATF-3)
  ↓ extends
ATF-PGL-Compliant  (+ RFC-ATF-4)  ← fourth tier
  ↓ extends
ATF-CGL-Compliant  (+ RFC-ATF-5)  ← fifth tier [RFC-ATF-5]
```

Each tier subsumes all lower tiers. An ATF-PGL-Compliant implementation is necessarily ATF-FEI-Compliant, ATF-RGC-Compliant, and ATF-Compliant. RFC-ATF-5 defines the fifth and highest tier, ATF-CGL-Compliant, which subsumes ATF-PGL-Compliant.

12. Implementation Requirements

The following requirements are normative. The key words MUST, MUST NOT, SHOULD, and MAY are as defined in BCP 14.

12.1. AGVP Requirements

REQ-AGVP-001 (MUST):

The AGVPWatchdog MUST be deployed as a persistent daemon thread with `AGVP_WATCHDOG_INTERVAL_SECONDS` ≥ 30 .

REQ-AGVP-002 (MUST):

PVRs MUST be persisted to a durable store (database) before the watchdog loop logs `ANTICIPATORY_VETO_ISSUED`.

REQ-AGVP-003 (MUST):

PVR identifiers MUST be globally unique. A database uniqueness constraint on `pvr_id` MUST be present to prevent duplicate issuance under concurrent execution.

REQ-AGVP-004 (MUST):

Governance request processing MUST check for ACTIVE PVRs and block accordingly (AGV-INV-001). The check MUST occur before any governance evaluation is performed.

REQ-AGVP-005 (MUST):

The AGVPWatchdog MUST NOT emit PVRs for domains where `AVM.evaluate()` returns `pass_through = True` (AGV-INV-005).

REQ-AGVP-006 (MUST NOT):

The AGVPWatchdog MUST NOT set PVR.status = REVOKED under any condition (AGV-INV-002).

REQ-AGVP-007 (SHOULD):

AGVP_WATCHDOG_INTERVAL_SECONDS SHOULD be configurable via environment variable. The default value SHOULD be 60 seconds.

REQ-AGVP-008 (MUST):

PVR content_hash MUST be computed as SHA-256 of canonical JSON of committed fields with keys sorted lexicographically.

REQ-AGVP-009 (MUST):

PVR pqc_algorithm MUST be "ML-DSA-65". Any other algorithm MUST be rejected at PVR construction time.

12.2. SSD Requirements

REQ-SSD-001 (MUST):

SSD.classify() MUST be called on every AVM.evaluate() invocation. StructuralShiftReport MUST be present in every AVMResult.

REQ-SSD-002 (MUST):

Auto-recalibration MUST check SSD.classify() before execution. If shift_class = STRUCTURAL_SHIFT, recalibration MUST be skipped and the skip MUST be logged at WARNING level.

REQ-SSD-003 (MUST):

The component history ring buffer MUST use FIFO eviction with a maximum of SSD_MAX_HISTORY (20) entries. Selective deletion is not permitted.

REQ-SSD-004 (MUST):

STRUCTURAL_SHIFT MUST NOT be returned when cycles_analyzed < 5. Low-history domains MUST return INSUFFICIENT_DATA.

REQ-SSD-005 (SHOULD):

SSD_TOP_K_COMPONENTS SHOULD be 3. Implementations MAY configure a different value, but the value MUST be at least 2.

REQ-SSD-006 (MUST):

If SSD.classify() raises any exception, the exception MUST be caught. The caller MUST receive an AVMResult with shift_class = INSUFFICIENT_DATA rather than an unhandled exception.

12.3. DSPP Requirements

REQ-DSPP-001 (MUST):

A TSA MUST be created and signed for every Delegation Receipt at issuance time. Failure to create a TSA MUST NOT prevent the receipt from being issued.

REQ-DSPP-002 (MUST):

TSAs MUST be publicly accessible without authentication. TSA retrieval MUST NOT require live access to the issuing runtime.

REQ-DSPP-003 (MUST):

SDRs MUST be published whenever the runtime's SPV changes. SDR publication MUST include a non-empty change_rationale.

REQ-DSPP-004 (MUST NOT):

SDR records MUST NOT be modified or deleted after publication. The SDR chain for any runtime MUST be append-only.

REQ-DSPP-005 (MUST):

RSA computation MUST be executable without any network call to the originating runtime.

REQ-DSPP-006 (MUST NOT):

The SDU weight constants (0.40, 0.35, 0.25) MUST NOT be configurable at runtime. Any attempt to override them MUST be rejected with a configuration error.

REQ-DSPP-007 (MUST NOT):

The DSPP portability verdict thresholds (0.10, 0.40, 0.70) MUST NOT be configurable at runtime.

REQ-DSPP-008 (MUST):

A SEMANTICALLY_INCOMPATIBLE verdict on a DR MUST be propagated to all descendants of that DR in the assessment result.

12.4. Formal Verification Requirements

REQ-FVS-001 (MUST):

ATF-PGL-Compliant implementations MUST include a machine-executable formal proof runner that reproduces the 19 UNSAT results from §8.2.

REQ-FVS-002 (SHOULD):

The proof runner SHOULD produce a machine-readable JSON report (§8.4 format) for inclusion in compliance deliverables.

REQ-FVS-003 (MUST NOT):

A proof that returns SAT or UNKNOWN for any invariant listed in §8.2 MUST NOT be used to claim ATF-PGL-Compliant status for that invariant. A SAT result means the invariant does not hold as stated.

13. Security Considerations

13.1. ProactiveVetoReceipt Spoofing

Threat: an adversary with write access to the `avm_anticipatory_veto_receipts` table may insert PVR records with fabricated `drift_scores` and `content_hashes`, creating fraudulent blocks for arbitrary domains.

Mitigation: PVR `content_hash` is SHA-256 of committed fields.

Any inserted PVR that does not carry a valid ML-DSA-65 signature from the platform signing key fails verification – the signature cannot be forged without the private key. ATF-PGL-Compliant implementations MUST verify PVR signatures before enforcing blocks (REQ-AGVP-004). The ML-DSA-65 private key MUST be protected under the same controls as the platform signing key (RFC-ATF-3 §14.6).

13.2. Semantic Anchor Replay Attack

Threat: an adversary publishes a fraudulent TSA that misrepresents the originating runtime's posture at issuance time – causing RSA computations to understate divergence (SEMANTICALLY_PORTABLE verdict for a receipt that should be DRIFT_CRITICAL).

Mitigation: TSA authenticity depends on the ML-DSA-65 signature from the issuing runtime's key. A receiving domain performing RSA MUST verify the TSA signature before consuming its `semantic_term_registry`. TSA signing uses the same platform signing key as Delegation Receipts; the public key is independently published (as per RFC-ATF-3 §10.1). A TSA with an invalid signature MUST cause the RSA to return an ASSESSMENT_FAILED result rather than a portability verdict.

13.3. CRSI History Flooding

Threat: an adversary with the ability to trigger frequent governance evaluations (through high-frequency agent requests) may attempt to flood the SSD ring buffer with entries that establish a false "stable" topology, masking an actual structural shift by overwriting the history buffer faster than topology changes can accumulate.

Mitigation: the ring buffer has a maximum of `SSD_MAX_HISTORY` (20) entries. At 1 request per second, a 20-entry buffer fills in 20 seconds – which is also the minimum useful SSD observation window. Implementations MAY rate-limit the frequency at which `AVM.evaluate()` contributes to SSD history (e.g., at most one history entry per N seconds per domain). Rate limiting the governance request path (RFC-ATF-3 §10.3) provides upstream mitigation.

13.4. Watchdog Starvation

Threat: if the request path is never executed for a blocked domain, the cached signal telemetry for that domain becomes stale. The watchdog continues to see stale signals and may never observe drift recovery – holding the ACTIVE PVR indefinitely.

Mitigation: the Signal Telemetry Update Protocol (§5.4) ensures that blocked requests still update cached signals – the update occurs before any block check. Even when every request is blocked, the signals are refreshed.

Implementations SHOULD configure a maximum signal staleness threshold consistent with the normative rule in §5.4 – signals older than $\max(3 \times \text{AGVP_WATCHDOG_INTERVAL_SECONDS}, 300 \text{ seconds})$ MUST NOT be used for new PVR issuance.

13.5. SDR Chain Forgery

Threat: an adversary with the ability to inject into the SDR chain for a runtime may attempt to add fraudulent SDR entries that show a MORE_RESTRICTIVE posture evolution – causing RSA computations to compute artificially low SDU values, making incompatible receipts appear portable.

Mitigation: SDR authenticity depends on ML-DSA-65 signatures. A receiving domain computing RSA MUST verify the signature of each SDR in the chain before consuming it. An SDR with an invalid signature MUST be treated as a chain break; the RSA MUST be flagged as UNVERIFIABLE for that chain segment.

13.6. Zero-Discount Delegation Cycle

Threat: by constructing a delegation cycle where each delegation uses `discount = 0` (no authority reduction), an adversary might attempt to construct an infinite delegation loop with equal budgets at every node.

Mitigation: ATF-INV-003 (RFC-ATF-1) establishes receipt immutability – a receipt cannot modify another receipt. The TLA+ AcyclicityInvariant (§8.3) proves that the ATF delegation graph has no directed cycle in any reachable state, regardless of discount value. The Z3 proof of RGC-INV-004 (§8.2) proves the strict-discount case arithmetically. The identity-based cycle prohibition (unique receipt IDs, `parent_id` cannot reference the receipt's own ID) prevents zero-discount cycles structurally.

14. Novel Contributions

RFC-ATF-4 introduces six novel technical contributions to the governance infrastructure literature.

14.1. Anticipatory Governance Veto Receipt (PVR)

– First Proactive Governance Attestation in AI Governance

No prior governance framework for autonomous AI agents emits a cryptographically signed, database-persisted veto artifact before a governance request arrives.

The PVR closes the forensic gap identified in §2.1. Its timestamp precedes any subsequent blocked request. The sequence:

```
[T=0:00 PVR signed and persisted – conditions were adverse HERE]
[T=4:00 request arrives – blocked, block_reason references PVR]
```

is forensically unambiguous. No prior specification – LangChain, AutoGPT, CrewAI, Microsoft AutoGen, or VeriSigil VGS-001/011 – provides an anticipatory governance receipt.

The PVR's structural position in the governance timeline (before requests) is what distinguishes it from enhanced reactive vetoes. It is not faster detection – it is categorically earlier attestation.

14.2. Component Rank Stability Index (CRSI)

– Novel Recalibration Safety Metric

CRSI is a position-weighted Jaccard overlap over the drift-component history – a continuous, formally bounded metric (Z3-proved: $CRSI \in [0,1]$ for all valid inputs) that classifies whether the signal topology driving governance drift is stable or has undergone structural change.

No prior work in AI governance distinguishes sustained excursion from topology change in drift analysis. The distinction is operationally critical: recalibrating to a sustained excursion is safe; recalibrating to a topology change embeds unvalidated assumptions as the new baseline.

The position-weighting design (§6.3) is itself novel: governance intuition about which components matter most (rank-1 = dominant driver) is formalized as a metric rather than left to operator judgment.

14.3. DSPP — O(1) Cross-Domain Assessment Without Negotiation

The DSPP RSA is a unilateral, offline computation that scales to arbitrarily many receiving domains with O(1) computation per receipt per domain. No contact with the originating runtime. No bilateral state machine. No per-domain negotiation.

The SDU formula (§7.2) is the first formally bounded semantic divergence metric defined for AI governance receipts. Its weight assignments reflect governance intuition: boundary conditions (0.40) matter more than scope (0.35), which matters more than regulatory anchors (0.25). The convex combination property (SDU-WSUM, Z3-proved) guarantees the metric remains normalized.

The protocol's scalability property is architecturally significant: an issuing runtime publishes a TSA once, and any number of receiving domains can assess portability at any future time without coordination.

14.4. Dual-Methodology Formal Verification (TLA+ + Z3 SMT)

– First in AI Governance

RFC-ATF-1 through RFC-ATF-3, combined with RFC-ATF-4, constitute the first AI governance protocol specification with:

- (a) TLA+ model-checked safety properties across all discrete state traces (MAR, Acyclicity, ChainRoot, Immutability, RCEpiry).
- (b) Z3 SMT arithmetic proofs for all arithmetic-defined invariants across the continuous input domain (19 proofs, all UNSAT).

The two methodologies cover the two classes of properties that governance specifications require: discrete state-machine safety (TLA+) and continuous arithmetic correctness (Z3). Using only one methodology leaves the other class unverified.

VeriSigil VGS-001/011 (the closest comparable published specification) provides 4 Z3 proofs and 5 TLA+ theorems but does not apply both methodologies to the same protocol. OMNIX FVS-1.0 provides 19 Z3 proofs over a protocol that also carries TLA+ coverage – 4.75× the Z3 proof count in a dual-methodology verified specification.

14.5. Proactive Governance Gap (Gap_PG) Formal Definition

– First Formal Gap Analysis for Anticipatory Governance

Section 2.4 introduces the Proactive Governance Gap as a formal definition:

$$\text{Gap_PG} = (\text{PVC} \cup \text{SP} \cup \text{TS}) - (\text{ICV} \cup \text{RVC})$$

This formalization enables compliance assessment to be stated precisely: a deployment has $\text{Gap_PG} = \emptyset$ if and only if it satisfies PVC (AGVP), SP (DSPP), and TS (SSD). No prior specification defines the gap between reactive and proactive governance coverage in formal terms.

14.6. ATF-PGL-Compliant — Fourth Compliance Tier

The ATF compliance hierarchy spans five tiers:

ATF-Compliant → ATF-RGC-Compliant → ATF-FEI-Compliant → ATF-PGL-Compliant → ATF-CGL-Compliant (RFC-ATF-5).

ATF-PGL-Compliant is the first compliance tier that addresses evidence between requests (AGVP), semantic legitimacy over time (DSPP), and recalibration safety (SSD) – the three proactive governance properties that no prior tier addressed. RFC-ATF-5 introduces ATF-CGL-Compliant, the fifth tier, addressing decision space documentation (CGE), universal multi-jurisdiction certification (GUGT), and longitudinal evidence interpretability (TGB).

15. Distinction from Prior Art

The following comparison covers the closest published governance specifications as of May 2026.

Property	OMNIX	VeriSigil	LangCh	AutoGen	OAuth
ATF	VGS				
Proactive veto before request	Yes	No	No	No	No
PVR: signed, timestamped	Yes	No	No	No	No
Recalibration topology check	Yes	No	No	No	No

■ CRSI: continuous metric	■ Yes	■ No	■ No	■ No	■ No	■
■ Semantic portability (0(1))	■ Yes	■ Partial*	■ No	■ No	■ No	■
■ No negotiation for portability	■ Yes	■ No	■ No	■ No	■ No	■
■ Z3 SMT proofs	■ 19	■ 4	■ 0	■ 0	■ 0	■
■ TLA+ properties	■ 5	■ 5	■ 0	■ 0	■ 0	■
■ Dual methodology (Z3 + TLA+)	■ Yes	■ No	■ No	■ No	■ No	■
■ Post-quantum signing (ML-DSA)	■ Yes	■ No	■ No	■ No	■ No	■
■ Offline-verifiable receipts	■ Yes	■ Yes	■ No	■ No	■ Yes	■
■ Formal compliance designation	■ Yes	■ Yes	■ No	■ No	■ Yes	■

* VeriSigil VGS-010 addresses "Jurisdiction-Aware Admissibility" but requires runtime-dependent assessment at verification time. DSPP RSA is computable entirely locally with no originating runtime contact. The portability model is categorically different.

OMNIX-specific properties shared by no other published specification:

- Anticipatory governance receipt (PVR) – signed, timestamped, persisted before any request arrives
- Component Rank Stability Index (CRSI) – topology stability metric
- DSPP Retroactive Semantic Assessment – $O(1)$, no negotiation
- Dual Z3 + TLA+ formal verification on the same protocol stack

16. Regulatory Alignment

Framework	AGVP Relevance	SSD Relevance	DSPP Relevance
EU AI Act Art. 9, 13	Art. 9 risk management: proactive risk detection	Art. 61 post-market: continuous recal. audit	Art. 13 transparency: cross-domain semantics
DORA Art. 6 (ICT risk)	Continuous monitoring with signed evidence	Topology change audit for recal. safety	–
DORA Art. 9 (resilience)	Detection latency $\leq 60s$	–	–
NIST AI RMF GOVERN 1.6	–	Recal. safety docs	Multi-domain semantics
NIST AI RMF MANAGE 4.1	Monitoring evidence	Topology assessment	–
China AI Law Arts. 20–22	–	–	Cross-border portability
GCC/DIFC AI Reg. 10	Sovereign attestation	–	Jurisdiction portability
SOC 2 Type II CC7.2	Continuous monitoring	Change documentation	–
ISO 27001 A.12.4	Signed monitoring trail	–	Semantic audit log
FATF Rec. 16 (AI finance)	Proactive freeze record	–	–

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18. Appendix A — AGVP Wire Format Reference

Normative field reference for a PVR JSON record
(avm_anticipatory_veto_receipts table schema):

Field	Type	Required	Constraints
pvr_id	string	REQUIRED	"PVR-" + 16 uppercase hex chars
tenant_id	string	REQUIRED	Non-empty
domain	string	REQUIRED	Non-empty
drift_score	number	REQUIRED	> drift_threshold; ≤ 1.0
drift_threshold	number	REQUIRED	> 0.0; < 1.0
drift_components	object	REQUIRED	{signal_name: float, ...}
structural_shift_class	string	REQUIRED	INSUFFICIENT_DATA STABLE DRIFT_WITH_INSTABILITY

				STRUCTURAL_SHIFT	
signals_at_assessment	object	REQUIRED	{signal_name: value, ...}		
assessment_timestamp	string	REQUIRED	ISO-8601-UTC, nanosecond precision		
veto_effective_from	string	REQUIRED	= assessment_timestamp		
status	string	REQUIRED	ACTIVE REVOKED		
content_hash	string	REQUIRED	"sha256:" + 64 lowercase hex chars		
pqc_signature	string	REQUIRED	ML-DSA-65 signature bytes, base64		
pqc_algorithm	string	REQUIRED	"ML-DSA-65"		
snapshot_id	string	OPTIONAL	Reference to active AVM snapshot		
created_at	string	REQUIRED	ISO-8601-UTC (DB insertion time)		

Committed fields for content_hash computation (canonical JSON, keys sorted lexicographically, no whitespace):
 pvr_id, tenant_id, domain, drift_score, drift_threshold,
 signals_at_assessment, assessment_timestamp, veto_effective_from,
 snapshot_id

19. Appendix B — SDU Computation Reference

SDU formula reference for all eight ATF Core Terms:

```
SDU(term_a, term_b):
  Step 1: Hash shortcut check
    if term_a.content_hash == term_b.content_hash: return 0.0

  Step 2: boundary_distance
    A = set(term_a.boundary_conditions)
    B = set(term_b.boundary_conditions)
    boundary_distance = |A ■ B| / |A ∪ B|
    (If A ∪ B is empty: boundary_distance = 0.0)

  Step 3: scope_distance
    C = set(term_a.operational_scope)
    D = set(term_b.operational_scope)
    scope_distance = 1.0 - |C ∩ D| / |C ∪ D|
    (If C ∪ D is empty: scope_distance = 0.0)

  Step 4: regulatory_distance
    E = set(term_a.regulatory_anchors)
    F = set(term_b.regulatory_anchors)
    regulatory_distance = |E ■ F| / |E ∪ F|
    (If E ∪ F is empty: regulatory_distance = 0.0)

  Step 5: SDU = 0.40 · boundary_distance
    + 0.35 · scope_distance
    + 0.25 · regulatory_distance
```

Portability verdict thresholds (structural constants,
 DSPP-INV-007a and DSPP-INV-007b):

Range	Verdict	Interpretation	
[0.00, 0.10)	SEMANTICALLY_PORTABLE	Use without qualification	
[0.10, 0.40)	DRIFT_ACKNOWLEDGED	Apply contextual qualification	
[0.40, 0.70)	DRIFT_CRITICAL	Document divergence; human review rec.	
[0.70, 1.00]	SEMANTICALLY_INCOMPAT.	Operational meaning unreliable in rcvr	

20. Appendix C — PGL Compliance Checklist

Implementations claiming ATF-PGL-Compliant MUST satisfy all items.
 Items marked (FEI) are inherited from RFC-ATF-3 and MUST be
 satisfied at that tier before this checklist applies.

AGVP

- AGVPWatchdog deployed with interval $\geq 30s$ (AGV-INV-003)
- PVRs persisted atomically; uniqueness constraint on pvr_id
- PVR content_hash = SHA-256(canonical_json(committed_fields))
- PVR pqc_algorithm = "ML-DSA-65" enforced at construction
- PVR signed with platform ML-DSA-65 key before persistence
- Request path checks has_active_pvr() before evaluation (AGV-INV-001)
- Blocked requests carry block_reason referencing PVR ID
- Signal telemetry updated by request path even when blocked (§5.4)
- Pass-through domains excluded from PVR emission (AGV-INV-005)
- Watchdog has no code path setting PVR.status = REVOKED (AGV-INV-002)
- Auto-recalibration checks has_active_pvr() before executing (AGV-INV-006)
- PVR records not deleted after revocation (audit trail preservation)

SSD

- SSD.classify() called on every AVM.evaluate() invocation
- StructuralShiftReport present in every AVMSResult
- StructuralShiftReport embedded in every PVR as structural_shift_class
- STRUCTURAL_SHIFT blocks auto-recalibration (SSD-INV-001)
- Component history: FIFO eviction only, max 20 entries (SSD-INV-002)
- STRUCTURAL_SHIFT only when cycles_analyzed ≥ 5 (SSD-INV-003)
- SSD exceptions caught; fallback = INSUFFICIENT_DATA (§9.3)

DSPP

- TSA created for every Delegation Receipt at issuance
- TSA signed with ML-DSA-65 and persisted
- TSAs publicly accessible without authentication (DSPP-INV-004)
- SDR published on every SPV change with non-empty rationale
- SDR chain is append-only; no modification or deletion (DSPP-INV-002)
- RSA computable with no originating runtime contact (DSPP-INV-004)
- SDU weights (0.40, 0.35, 0.25) are hardcoded constants (DSPP-INV-007a and DSPP-INV-007b)
- Verdict thresholds (0.10, 0.40, 0.70) are hardcoded constants
- SEMANTICALLY_INCOMPATIBLE propagates to DR descendants (DSPP-INV-006)
- TSA signing failure: receipt still issued, ERROR logged (§9.3)

Formal Verification

- Proof runner executes 19/19 proofs as UNSAT (§8.2)
- JSON proof report producible for compliance deliverables (§8.4)
- No SAT or UNKNOWN results for any invariant in §8.2 table

Inherited from ATF-FEI-Compliant (RFC-ATF-3) (abbreviated):

- (FEI) All RFC-ATF-1 invariants satisfied (ATF-INV-001-006)
- (FEI) All RFC-ATF-2 invariants satisfied (RGC-INV-001-008)
- (FEI) All RFC-ATF-3 invariants satisfied (GPIL/ELR/EAP/OEP/FEA/FVP)
- (FEI) PostgreSQL + Redis configured in production
- (FEI) ML-DSA-65 signing keys stable across restarts (FMR-001)
- (FEI) OMNIX_ANTI_REPLAY_MODE = strict in production

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